Curiosity-driven Exploration by Self-supervised Prediction

Introduction

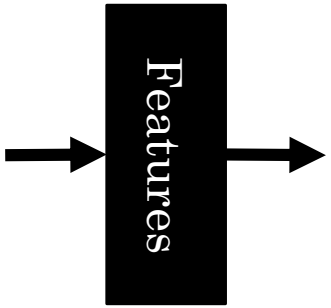
- Extrinsic rewards are extremely sparse; curiosity can serve as an intrinsic reward signal.
- We formulate curiosity as the error in an agent's ability to predict the consequence of its own actions.
- We use visual feature space learned by a self-supervised inverse dynamics model.

Why use visual feature space?

- Measuring "novelty" requires a model of environmental dynamics that predicts the next state. The model is hard to build in high-dimensional situations such as images.
- Stochasticity (e.g. noise) of the agent-environment system causes to move in a stochastic manner.
- We only predict those changes in the environment that could possibly be due to the actions of our agent or affect the agent, and ignore the rest.

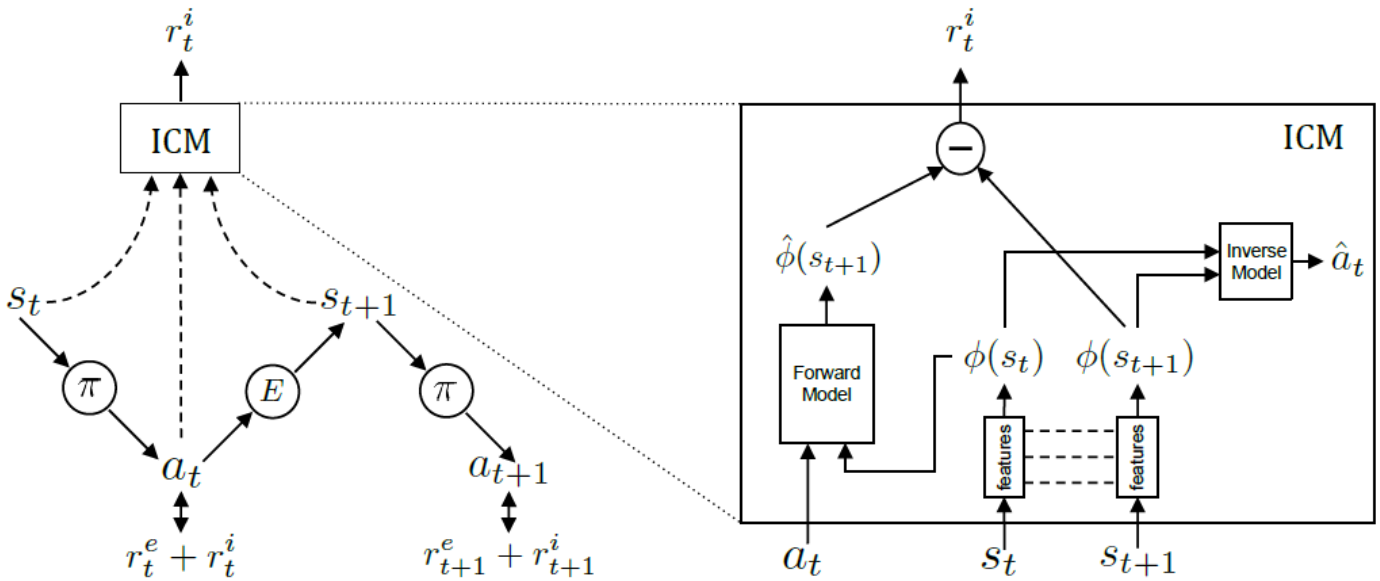
$S_t =$

(Raw pixel space)



$\phi(S_t)$

(Feature space)



- Policy π is trained to optimize the sum of rewards

$$\max_{\theta_P} \mathbb{E}_{\pi(s_t;\theta_P)}[\Sigma_t r_t]$$

- ICM (Intrinsic Curiosity Module) encodes the states into features $\phi(s_t)$ that are trained to predict a_t.
- The prediction error in the feature space is used as intrinsic reward.

Inverse Dynamics Model

- This is a mechanism to learn feature representations.
- Such a feature space can be learned by training a deep neural network with two sub-modules:
- The first sub-module encodes the raw state S_t into a feature vector $\phi(S_t)$
- The second sub-module takes the feature encodings $\phi(S_t),\ \phi(S_{t+1})$ and predicts the action a_t
- Training this neural network amounts to learning function g, AKA Inverse Dynamics Model.

$$\hat{a}_t = g\left(s_t, s_{t+1}; \theta_I\right)$$

$$\min_{\theta_I} L_I(\hat{a}_t, a_t)$$

Forward Dynamics Model

- We train another neural network that takes a_t and $\phi(S_t)$ as inputs and predicts the feature encoding of the next state.

$$\hat{\phi}(s_{t+1}) = f\Big(\phi(s_t), a_t; \theta_F\Big)$$

$$L_F\Big(\phi(s_t), \hat{\phi}(s_{t+1})\Big) = \frac{1}{2}\|\hat{\phi}(s_{t+1}) - \phi(s_{t+1})\|_2^2$$

- The intrinsic signal is computed as

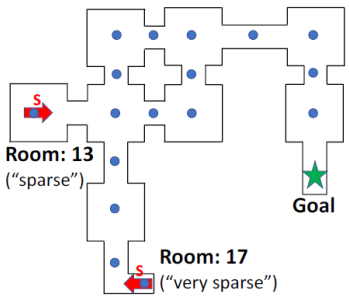
$$r_t^i = \frac{\eta}{2}\|\hat{\phi}(s_{t+1}) - \phi(s_{t+1})\|_2^2$$

- The overall optimization problem

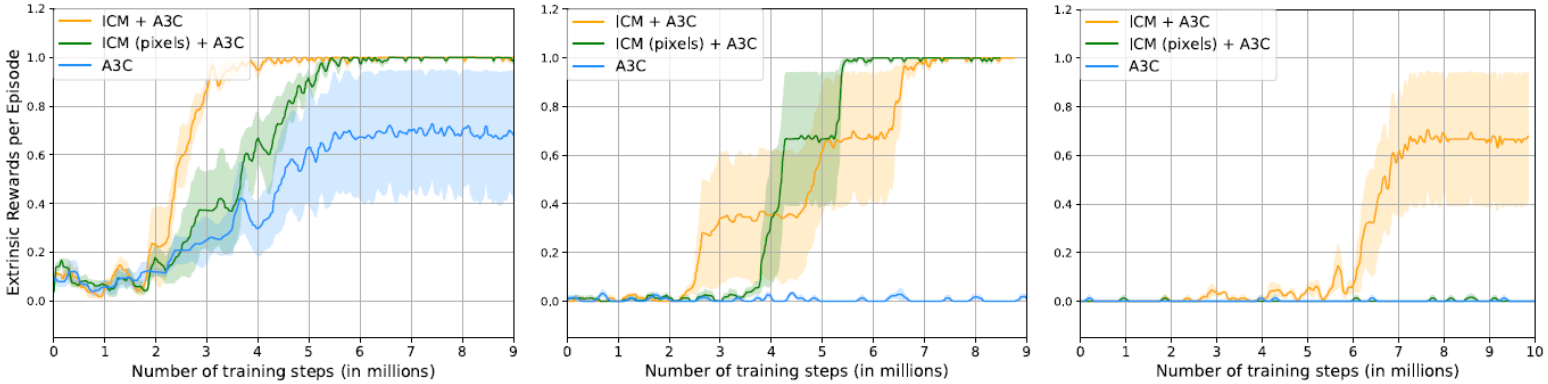
$$\min_{\theta_P, \theta_I, \theta_F} \Big[-\lambda \mathbb{E}_{\pi(s_t;\theta_P)}[\Sigma_t r_t] + (1-\beta)L_I + \beta L_F \Big]$$

Varying degree of sparsity

- ICM is better suited than ICM-pixels and vanilla A3C for hard goal directed exploration tasks.



(b) Test Map Scenario

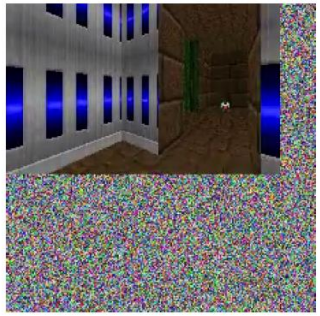


(a) "dense reward" setting (b) "sparse reward" setting (c) "very sparse reward" setting

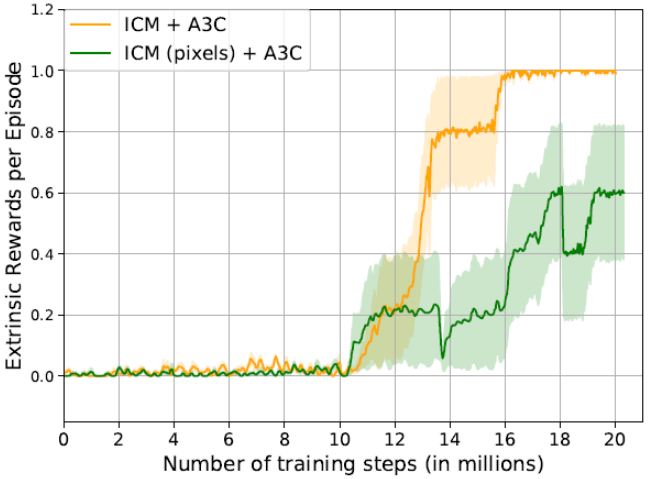
Robust to uncontrollable dynamics

- We augmented the agent's observation with a fixed region of white noise.
- This indicates that ICM is insensitive to nuisance changes in the environment.

(a) Input snapshot in VizDoom



(b) Input w/ noise



Comparison to TRPO-VIME

- VizDoom "sparse" reward setup
- ICM agents work significantly better than TRPO and TRPO-VIME, both in terms of convergence rate and accuracy.

Method ("sparse" reward setup)	Mean (Median) Score (at convergence)
TRPO	26.0 % (0.0 %)
A3C	0.0 % (0.0 %)
VIME + TRPO	46.1 % (27.1 %)
ICM + A3C	**100.0** % (100.0 %)